

LLM-Mediated Nudge-Based Text Detoxification: Influencing User Choices to Mitigate Hate Speech

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Abstract—This position paper presents a proposal for an LLM-mediated Nudge-Based Text Detoxification framework, designed to proactively mitigate hate speech without resorting to coercive interventions or automatic text rewriting. To address the limitations observed in conventional Text Detoxification approaches, often associated with user passivity and the perception of algorithmic censorship, this work aims to reframe text detoxification as an assisted decision-making process, in which the system acts as a cognitive mediator. Methodologically, the study adopts the Design Science Research approach to ground the formulation of behavioral conjectures and the development of a conceptual framework that integrates toxicity analysis and principles of Digital Nudging and Human-Computer Interaction. Furthermore, the article describes an exploratory empirical evaluation strategy, combining quantitative and qualitative methods, aimed at investigating the artifact's acceptability and the plausibility of the conjectures guiding its design. Through the proposed framework and the designed experimental procedure, future evaluations are expected to provide initial evidence that nudging-based interventions can increase the acceptance of Text Detoxification systems and reduce the perception of algorithmic censorship, preserving autonomy and perceived freedom of choice, while contributing to the design of user-centered text detoxification systems.

Index Terms—Hate Speech, Text Detoxification, Digital Nudges, Cognitive Interventions, Large Language Models

I. INTRODUCTION

Hate speech and other forms of toxic language constitute one of the main contemporary challenges for digital platforms [1]. In response, automated content detection and moderation systems have been widely adopted, especially those based on machine learning and Large Language Models (LLMs) [2]. Despite technical advances, many of these solutions still rely on reactive approaches or automatic interventions that assign the user a passive role, which has been associated with perceptions of algorithmic censorship, loss of autonomy,

and low social acceptability [3]. In this context, proactive moderation strategies, such as Text Detoxification (TD), seek to intervene at the moment of writing the hateful text [4], but recent reviews indicate that such approaches still neglect fundamental Human-Computer Interaction (HCI) principles, such as transparency, user control, and freedom of choice [5].

This work is based on the premise that TD can be reinterpreted as an assisted decision-making process, in which the system acts as a cognitive mediator rather than a coercive agent. To this end, the work incorporates interventions based on Nudge Theory and its extension in the digital context, Digital Nudging [6]. The main objective of this article is to propose and substantiate an LLM-mediated framework for Nudge-Based Text Detoxification, guided by HCI principles and conceived according to the Design Science Research approach, specifically the DSR-Model [7]. The research contributions consist of a conceptual reframing of TD as assisted decision-making, the proposition of a framework that systematizes design principles and guidelines for user-centered text detoxification systems, and the definition of an empirical evaluation strategy aligned with the DSR-Model.

In accordance with the DSR-Model, the article is structured according to its main elements. Section II presents the knowledge about the problem and the context, including relevant literature on TD, proactive moderation, and digital nudging. Section III explicitly states the behavioral conjectures that underpin the artifact design. Section IV describes the proposed framework and its conceptual components. Section V presents the empirical evaluation strategy of the artifact. Finally, Sections VI and VII respectively discuss the expected theoretical findings and the conclusions of this work.

II. KNOWLEDGE ABOUT THE PROBLEM AND CONTEXT

Hate speech is defined as the use of language that attacks individuals or groups based on characteristics such as nationality, ethnicity, religion, sexual orientation, and others [1]. In this regard, two main approaches are used to moderate hateful content: reactive and proactive [3]. Reactive approaches focus on addressing hate speech after it has been generated; in this sense, strategies such as account suspension or even content deletion based on rules defined by social networks are prominent [8]. Such approaches present significant limitations, especially regarding scalability, the perception of censorship, and low effectiveness in preventing the recurrence of toxic behaviors.

In contrast, proactive approaches concentrate on preventing the production of toxic content, acting before the moment of publication [9]. These approaches involve interventions during the writing process, encouraging users to reconsider or rephrase potentially harmful messages. A central differentiator of proactive strategies is their educational and reflective potential, as they can contribute to the development of critical awareness on the part of the user and to the reduction of hate speech recurrence over time [10].

In this context, one of the most prominent proactive approaches is Text Detoxification (TD), which is defined as the process of automatically rewriting toxic text to remove offensive elements while preserving its semantic intent [11]. The most common approaches include fine-tuning sequential models, prompting techniques with LLMs, and the use of penalties based on toxicity classifiers [2, 5].

Despite showing promising technical results in various scenarios, TD approaches share a central limitation: the predominantly passive role assigned to the user. In general, the system unilaterally determines what is considered to be acceptable, how the text should be modified, and when the modification occurs, often without offering transparency regarding the process or the possibility of choice to the user [3]. This characteristic can generate resistance, a perception of algorithmic censorship, and questions regarding autonomy and freedom of expression, constituting a sociotechnical problem that has received scant attention in the literature to date.

A. Problem in Context

Hate speech on digital platforms constitutes a complex sociotechnical problem, whose individual and collective impacts have been widely documented in the literature [1]. In response, digital platforms have been adopting automated moderation systems, predominantly grounded in reactive approaches of content detection and removal, as well as sanctions applied to users who violate established policies [12]. More recently, TD has emerged as a proactive alternative, proposing the rewriting of potentially toxic texts into linguistically less offensive versions.

Furthermore, it is observed that the literature lacks proposals that systematize the design and implementation process of TD systems under a user-centered perspective. In general, existing solutions focus on algorithmic aspects and performance

metrics, neglecting fundamental principles such as autonomy, transparency, and informed choice [9]. This gap highlights the need for guidelines or models that guide the design of text detoxification systems not only from a technical point of view but also from cognitive and interaction perspectives.

In this context, the need emerges to formalize a process for the construction of TD systems that incorporates cognitive interventions capable of influencing linguistic choices without resorting to coercive mechanisms [10]. Such a process should seek to reduce the user's sense of passivity and mitigate concerns related to freedom of expression, repositioning text detoxification as a support for decision-making and reflection.

B. Theoretical Background

The theoretical background of this research is built upon the articulation of three complementary axes: Nudge Theory and its developments in the digital context, HCI principles applied to sensitive sociotechnical systems, and Natural Language Processing techniques based on LLMs applied to the hate speech context. This articulation enables establishing a relationship between theoretical knowledge about human behavior and decision-making and technical knowledge related to the design and implementation of the artifact proposed in this article, as established by the utilized Design Science Research approach [7].

Nudge Theory, introduced in [13], starts from the premise that human decisions are strongly influenced by how options are presented in the choice environment. Nudges are subtle interventions that alter people's behavior in a predictable way, without restricting options or imposing significant economic or legal penalties. Central elements of this theory include the notion of choice architecture, default option effects, framing effects, and reflective nudges, which seek to promote greater awareness and deliberation by the individual.

In the context of digital systems, these principles were systematized under the concept of Digital Nudging [6], which refers to the intentional use of interface elements, interaction flows, and computational mechanisms to influence user decisions in digital environments. Unlike nudges in physical contexts, digital nudging allows for a high degree of personalization, scalability, and dynamic adaptation, characteristics particularly relevant in natural language-based systems.

Applied to the TD problem, Nudge Theory offers a conceptual basis for reinterpreting hate speech mitigation not as a process of automatic content correction, but as an assisted decision-making process. Instead of replacing the user's text, the system acts as a persuasive mediator, presenting structured and transparent alternatives that encourage the choice of less toxic versions, preserving autonomy and freedom of expression. This reframing is especially relevant in light of recent criticisms of automatic rewriting, which point to risks of algorithmic censorship, loss of user control, and rejection of interventions [3].

From an HCI perspective, this approach aligns with user-centered design principles, algorithmic transparency, and explainability, which are fundamental in sociotechnical systems

dealing with sensitive values such as speech, identity, and expression [14]. By incorporating Nudge Theory concepts, such as reflective nudges, pre-selected default response effects, and framing strategies for textual alternatives to hateful text, the system seeks not only to reduce content toxicity but also to promote user reflection and learning, approaching proactive and participatory moderation models.

From a technical point of view, LLMs enable this approach by allowing both contextualized toxicity analysis and the generation of multiple textual alternatives that preserve the original semantic intent [2], but without the hateful content. Unlike traditional approaches based on lexical substitutions or fixed rules, LLMs enable stylistic, pragmatic, and discursive variations, which is essential for implementing rich and flexible choice architectures in the context of TD.

C. State of the Art

The literature on hate speech detection and moderation highlights the conceptual complexity of the phenomenon itself, whose definition and manifestation vary according to socio-cultural context, local norms, and pragmatic interpretations of language [1]. This characteristic makes the adoption of fully automated approaches particularly challenging, as they may fail to capture discursive nuances, reinforce existing biases, or produce interventions perceived as unfair or arbitrary.

Much of the research in the area has focused predominantly on algorithmic aspects of hate speech detection and mitigation, prioritizing performance metrics such as accuracy and semantic preservation. In this context, classical TD approaches explore euphemistic substitutions, penalties based on toxicity classifiers, and automatic rewriting techniques [5]. More recently, methods based on LLMs have demonstrated significant advances in generating less offensive texts, with greater linguistic naturalness and better preservation of the original meaning [2].

Despite these technical advances, recent studies point to important limitations regarding the user's absence in the decision-making process. As discussed in [3], the majority of TD solutions adopt a predominantly automatic stance, in which the system decides when and how the text should be modified, without the active involvement of the message author.

Some more recent works have begun to explore the behavioral dimension of content moderation. In the study presented in [15], concepts of persuasive technologies are combined with LLMs not only for the identification of cyberhate content but also to influence user behavior toward more sociable communication practices. Nevertheless, the authors themselves acknowledge the scarcity of systematic investigations addressing user interaction with these interventions, as well as their effects on perceived control, acceptance, and behavioral change.

Thus, although there are significant advances in the detection and automatic rewriting of toxic language, a relevant gap is observed in the state of the art regarding the formalization of processes and guidelines for the design of user-centered TD systems. In particular, there is a lack of guidelines that

explicitly integrate technical, cognitive, and interaction aspects, offering systematic guidance for the construction of non-coercive interventions that mitigate hate speech without intensifying perceptions of censorship or restriction of freedom of expression. It is in this space that the present research is situated.

III. BEHAVIORAL CONJECTURES

In line with the DSR-Model, the design of the proposed artifact is guided by a set of behavioral conjectures that express grounded assumptions about how users interact with TD systems and how such interactions can influence their linguistic decisions. These conjectures describe expected regularities in human behavior and bridge the gap between the theoretical background and the artifact's design decisions, serving as a basis for formulating the research questions and hypotheses to be investigated empirically. The behavioral conjectures guiding this research are as follows:

- **C1:** People tend to exhibit resistance to systems that automatically modify their texts without clear explanation of the process or without the possibility of choice by the user [3].
- **C2:** People are more likely to accept textual reformulation suggestions when they perceive they maintain explicit control over the final decision regarding the content to be published [3].
- **C3:** In contexts of potentially toxic but not prohibited language, people tend to select less toxic linguistic alternatives when these are presented as default options, provided that the original version remains accessible and the intervention is perceived as non-coercive [13].
- **C4:** People tend to reflect more on the social impact of their language when exposed to reflective, transparent, and contextualized nudges at the moment of writing [10].

These conjectures play a central role in the design of the artifact proposed in this research. Furthermore, they constitute behavioral hypotheses that will be evaluated through empirical investigation of the artifact's usage, enabling the production of scientific knowledge regarding the effects of LLM-mediated digital nudges on hate speech mitigation.

IV. A FRAMEWORK FOR LLM-MEDIATED NUDGE-BASED TEXT DETOXIFICATION

The core artifact of this research consists of a framework for Nudge-Based Text Detoxification mediated by LLM. The framework is proposed as the solution to the problems identified in the context of proactive hate speech moderation, especially user passivity, the perception of coercion, and the absence of systematic guidelines for the design of user-centered TD systems.

Unlike solutions limited to automatic text rewriting, the framework proposes a set of principles, conceptual components, and design guidelines that guide the construction of TD systems based on non-coercive interventions. Its objective is to influence linguistic choices transparently, preserving autonomy

and perceived freedom of expression, while promoting reflection and awareness regarding the social impacts of language.

The framework was conceived to explicitly materialize the behavioral conjectures presented in Section III, serving simultaneously as a prescriptive artifact to guide the development of text detoxification systems and as an analytical instrument to empirically investigate how LLM-mediated digital nudges influence user behavior in sensitive contexts.

A. Artifact Development Methodology

The development of the framework follows the iterative and reflective logic of DSR-Model, in which the artifact is progressively built, analyzed, and refined in light of behavioral theories and empirical evidence [7]. Initially, the problem in context, characterized by the limitations of traditional TD approaches regarding autonomy, transparency, and acceptance, grounds the formulation of the behavioral conjectures. These conjectures, in turn, guide design decisions at a conceptual level, defining which types of intervention are appropriate, when they should be applied, and how they should be presented to the user.

The framework adopts a modular and abstract structure, allowing for multiple implementation instances. This characteristic is essential to avoid coupling the artifact to a specific technical solution and to enable experimentation with different nudging strategies, choice architectures, and forms of interaction. Furthermore, modularity allows for addressing long-term behavioral issues, such as desensitization to repetitive nudges and the need for variation and adaptation of interventions to preserve their effectiveness.

B. Framework Overview

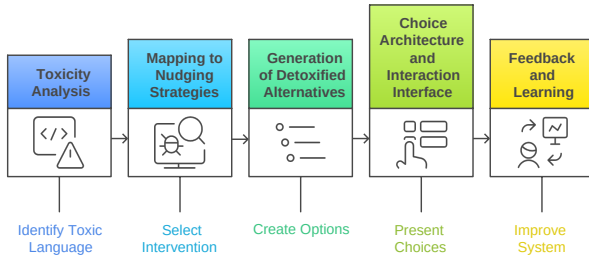


Fig. 1. LLM-Mediated Nudge-Based Text Detoxification framework components.

The framework is composed of five conceptual components that structure the Nudge-Based Text Detoxification process as seen in Fig. 1. These components should not be interpreted merely as technical modules, but as elements of a sociotechnical process that articulates automated analysis, cognitive intervention, human decision-making, and system learning. Together, they offer directions for designing systems that reduce textual toxicity while preserving user autonomy and perceived freedom of choice.

C. Component 1 – Toxicity Analysis

The Toxicity Analysis Component is responsible for identifying whether the text produced by the user presents indications of potentially toxic language and for determining the nature and severity of this language, to adequately guide the type of intervention to be performed. Because toxicity is not homogeneous, this component plays a fundamental role in distinguishing between cases amenable to mitigation through nudging and situations requiring other types of handling.

By implementing this component using LLMs, it is possible to address the following functions in a more specific and contextualized manner: detection of the presence of toxicity, classification of the toxicity type, identification of the target of offensive language, and extraction of problematic segments. Furthermore, the framework proposes that the analysis include an estimate of the degree of speech severity. From an implementation perspective, the LLM is operationalized as an analytical classifier using zero-shot or few-shot prompting to perform these multi-label toxicity detections and context extractions without relying on rigid rules [3].

In cases of potentially toxic, but not explicitly prohibited language, the system can proceed with the application of interventions based on nudging, as defined in subsequent components. However, in situations of extreme hate speech, such as explicit incitement to violence, advocacy of extermination practices, or systematic dehumanization of groups, the framework establishes that such messages should not be treated as simple candidates for linguistic softening. In these cases, the application of reformulation nudges may be inappropriate or ethically questionable, as it runs the risk of normalizing or legitimizing gravely harmful content.

However, it is imperative to acknowledge that the boundary separating moderate toxicity that is amenable to nudging from extreme hate speech is highly subjective. As highlighted in the literature, the definition and interpretation of toxic language vary significantly according to sociocultural context, local norms, and pragmatic nuances [1]. Consequently, establishing a rigid automated threshold presents significant challenges, as algorithms may struggle to capture these contextual subtleties.

Thus, the Toxicity Analysis Component also acts as a normative safeguard mechanism. Given the inherent subjectivity of this classification, the framework proposes that ambiguous cases, or those bordering extreme toxicity, should err on the side of caution. In such instances, the component signals that cognitive intervention based on nudging is not appropriate and that other strategies, such as blocking publication, referral to human moderation, or application of platform policies, should be considered. This distinction is essential to preserve the legitimacy of the framework, preventing it from being perceived as an instrument for the relativization of severe hate speech and reinforcing its commitment to responsible and ethically grounded moderation.

D. Component 2 – Mapping to Nudging Strategies

Once the presence of potentially toxic language is identified, the Mapping Component establishes how the system should

intervene, associating different toxicity types with specific nudging strategies. This component plays a central role in the framework, as it is responsible for translating the linguistic analysis performed in the previous component into cognitive interventions deliberately designed to influence linguistic decisions, without resorting to coercive or restrictive mechanisms.

The mapping between language types and nudging strategies, supported by LLMs, allows for greater adaptability to the message context, user interaction history, and behavioral recurrence. Technically, the LLM is operationalized as a contextual reasoning engine that receives the toxicity profile from the first component and leverages in-context learning to dynamically map the toxic speech to the most appropriate cognitive intervention. The framework establishes conceptual guidelines that guide this mapping according to the principle of minimum intervention, in order to reduce risks associated with cognitive load, desensitization to repetitive nudges, and the perception of censorship or manipulation.

In situations characterized by direct insults or explicitly aggressive language, the framework provides for the use of non-coercive default alternatives, in which a linguistically softened version is presented as a pre-selected option, while keeping the original version accessible. This strategy explores the default effect to influence choices subtly, preserving user autonomy and addressing the conjecture C3 that default options tend to be accepted when they do not imply a perceived loss of control.

In the case of offensive generalizations or stereotypes, the framework recommends interventions based on comparative rewrites, which preserve the critical intent of the original text but remove absolute or generalizing formulations. By presenting alternatives that highlight subtle differences in tone and framing, the system allows the user to explicitly perceive how small linguistic alterations can modify the message's social impact, promoting implicit learning and gradual reflection.

In ambiguous contexts, such as the use of sarcasm or irony with offensive potential, the framework proposes the adoption of reflective nudges, presented in the form of short questions or explanatory messages. These interventions seek to interrupt impulsive responses and stimulate conscious reflection at the moment of writing, calling attention to possible interpretations by third parties or unintended emotional impacts of the message.

For cases of veiled speech or potentially discriminatory language that is not explicit, the framework contemplates brief educational interventions, formulated in a non-normative and non-accusatory manner. These messages can provide contextual information regarding the sensitive nature of certain terms or expressions, favoring the gradual construction of awareness by the user, without resorting to moral judgments or implicit sanctions.

Furthermore, the Mapping Component incorporates guidelines for the variation and adaptation of nudging strategies over time, as a way to mitigate the risk of desensitization resulting from repetitive interventions. The alternation between different strategies and adaptation based on the user's choice history

allows preserving the system's effectiveness and reducing the perception of insistence or undue manipulation.

By systematizing these guidelines, the Mapping Component transforms abstract concepts from Nudge Theory into operational principles for the design of TD systems. Moreover, it enables the comparative empirical investigation of the effects of different nudging strategies, especially regarding intervention acceptance and the preservation of perceived freedom of expression, contributing directly to the validation of conjecture C3.

E. Component 3 – Generation of Detoxified Alternatives

The Generation Module utilizes LLMs to produce multiple alternative versions of the original text. The alternatives are generated with the objective of preserving the user's semantic intent, removing or softening offensive terms, and offering stylistic and pragmatic variations of the discourse.

The Generation Module utilizes LLMs to produce multiple alternative versions of the original text. In terms of operationalization, the LLM applies style-transfer techniques via targeted prompts to produce these detoxified alternatives. The alternatives are generated with the objective of preserving the user's semantic intent, removing or softening offensive terms, and offering stylistic and pragmatic variations of the discourse.

The generation of multiple alternatives is a central element for the framework's choice architecture, as it enables the simultaneous presentation of the original text and less toxic versions, ensuring that the user maintains control over the final decision. Thus, this module contributes directly to addressing conjectures C1 and C2, by avoiding automatic rewrites and preserving user autonomy.

This component guides the generation of multiple textual alternatives that preserve the user's communicative intent, while simultaneously reducing offensive elements. The existence of explicit alternatives, rather than automatic substitutions, is central to maintaining the user as the decision-making agent. This component reinforces the system's role as a mediator and not as a source of censorship, contributing directly to mitigating the sense of loss of freedom of expression.

F. Component 4 – Choice Architecture and Interaction Interface

This module defines how the generated alternatives are presented to the user. Inspired by HCI and Digital Nudging principles, this module encompasses elements such as the side-by-side presentation of textual versions, visualization of differences between the original text and the alternatives, and subtle indication of recommendations, such as the definition of a non-coercive default option.

This module is primarily responsible for materializing conjectures C2, C3, and C4 by structuring the decision environment in a way that influences linguistic choices without eliminating options. Furthermore, the interface may incorporate reflective nudges, such as questions or short messages that encourage the user to reflect on the social impact of their language at the moment of writing.

G. Component 5 – Feedback and Learning

The Feedback and Learning Component establishes guidelines for the collection and responsible use of interaction data, not only to improve system performance but also to evaluate its cognitive and behavioral effects. Beyond acceptance and choice metrics, this component emphasizes the observation of signs of reflection, conscious rejection, and behavioral change over time. Thus, the framework promotes not only the immediate reduction of toxicity but also the gradual construction of user awareness regarding the social impact of their language.

Together, these components characterize the framework as a prescriptive and investigative solution. By articulating theoretical principles, design guidelines, and conceptual components, the artifact provides a systematic basis for the development and evaluation of TD systems that respect user autonomy and explore the potential of LLMs as linguistic decision mediators, rather than merely as automation mechanisms.

V. PROPOSED EMPIRICAL EVALUATION

As this work is presented as a position paper, the empirical evaluation of the framework is slated for future research. The evaluation strategy proposed here aims to investigate whether the proposed artifact is accepted by users and whether the behavioral conjectures that grounded its design prove to be plausible when observed in use. In line with the DSR-Model [7], the planned evaluation seeks not only to verify the artifact’s technical functioning but also to produce evidence regarding its behavioral and perceptual effects in realistic interaction contexts.

Given the sociotechnical complexity of the problem, a comprehensive empirical evaluation is proposed, based on the interaction of a broad cohort of real users with a functional prototype of the Nudge-Based Text Detoxification framework.

A. Methodological Approach

The empirical evaluation will be conducted through a controlled experimental study, exploratory in nature, combining quantitative and qualitative methods. The quantitative approach aims to observe behavioral patterns and measure artifact acceptance, while the qualitative approach seeks to understand users’ perceptions, interpretations, and subjective reactions to the proposed interventions.

This type of mixed-method is widely adopted in HCI research for evaluating interactive systems in early design stages, as it allows capturing both measurable evidence and experienced aspects of the interaction [16]. The combination of these approaches is particularly relevant in the context of this work, since the framework’s objective is not limited to the technical reduction of textual toxicity but involves dimensions such as perceived autonomy, sense of freedom of expression, and acceptance of digital nudges.

B. Experimental Procedure

A functional prototype based on the proposed framework will be developed, implementing its main modules: toxicity analysis, mapping to nudging strategies, generation of

detoxified alternatives, and choice architecture and learning. The prototype will allow users to interact with the system in individual, short-duration sessions held in a controlled environment.

The study will involve between 150 volunteers, a robust sample size capable of yielding statistically significant quantitative data regarding artifact acceptance, while still allowing for the identification of behavioral patterns and design implications. Participants will be invited to perform writing tasks in simulated scenarios in which they must draft messages related to potentially polarizing topics, such as controversial social, political, or cultural issues.

The deliberate choice of polarizing themes is justified by the fact that neutral or low-emotional involvement topics tend not to generate offensive or toxic language, which could compromise the observation of the framework’s proposed interventions. Thus, the scenarios are designed to increase the probability of potentially problematic language occurrence without explicitly inducing hateful behaviors, preserving the experiment’s ethical character.

During the interaction, whenever the system identifies potentially toxic language, different nudging strategies may be triggered, presenting detoxified alternatives to the user. In all cases, the original text version will remain accessible, allowing the observation of choices effectively made and the evaluation of the effects of the proposed choice architecture.

C. Data Collection

The empirical evaluation will involve the collection of quantitative and qualitative data, organized into three main categories. Behavioral data include the text selected by the user, the number of edits performed after the nudge presentation, the reflection time before the final decision, and the explicit rejection of suggestions offered by the system. These data will allow analyzing choice patterns and verifying the influence of nudging strategies, especially regarding the acceptance of default alternatives.

To provide methodological clarity, the collected data will be translated into specific evaluation metrics to explicitly test the behavioral conjectures outlined in Section III:

- **C1 (Resistance to unilateral changes):** Evaluated through the rejection rate of system suggestions and qualitative scores on perceived algorithmic censorship and loss of control.
- **C2 (Acceptance with explicit control):** Measured by the acceptance rate of suggested detoxified alternatives and subjective ratings of perceived autonomy derived from the extended Unified Theory of Acceptance and Use of Technology (UTAUT) [17].
- **C3 (Effectiveness of default options):** Tested by calculating the retention frequency of the default non-coercive detoxified options compared to the rate of manual reversions to the original toxic text.
- **C4 (Reflection promotion):** Assessed via behavioral metrics such as reflection time measured in seconds between the nudge presentation and the user’s final decision

and the volume of self-edits performed after exposure to reflective nudges.

This explicit mapping ensures that each theoretical conjecture is empirically validated through a combination of objective interaction logs and validated psychometric scales.

Acceptance data will be collected through questionnaires based on consolidated technology acceptance models, such as TAM (Technology Acceptance Model) and UTAUT [17], covering dimensions such as perceived usefulness, ease of use, sense of control, transparency, and future behavioral intention to use. These data will provide quantitative evidence regarding the artifact’s acceptability.

Furthermore, qualitative data will be obtained from open-ended questions and Likert-type scales related to the perception of coercion, preservation of freedom of expression, fairness of interventions, and perceived system utility. The qualitative analysis of these responses will allow understanding how users interpret the nudges and whether they perceive the system as decision support or as a form of censorship.

This empirical evaluation strategy enables, in a feasible and methodologically rigorous manner, the investigation of the proposed framework’s adequacy from both technical and behavioral perspectives. By observing the artifact’s use by real users, even on a limited scale, the research meets the DSR-Model requirements and produces empirical evidence relevant to the advancement of TD approaches based on nudging and LLMs.

VI. EXPECTED THEORETICAL FINDINGS

In line with Design Science Research principles, this research is not limited to the development of a technical artifact but seeks to produce scientific knowledge from the empirical investigation of its use in context. Thus, it is expected that the execution of the research results in a set of theoretical findings that contribute to the advancement of the conceptual framework on LLM-mediated hate speech mitigation, digital nudging, and HCI.

First, it is expected to contribute to the understanding of the user’s role in TD systems. By rethinking text detoxification as an assisted decision-making process rather than automatic correction, the research can provide evidence that nudging-based approaches are more compatible with principles of autonomy, transparency, and perceived acceptance, expanding the theoretical debate on proactive and user-centered moderation.

Furthermore, the research seeks to deepen the understanding of the application of classic Nudge Theory concepts in the context of writing mediated by intelligent systems. It is expected to identify which nudging strategies are most effective in influencing linguistic choices toward less toxic alternatives without resorting to coercive mechanisms. These findings may contribute to the digital nudging literature, expanding its scope to the domain of content moderation and sensitive language.

Additionally, it is expected to produce theoretical knowledge on how LLMs can act not only as text generators or editors but as sociotechnical mediators supporting reflective decision-making processes. In this sense, the research can offer an

alternative perspective to the predominant use of LLMs as automation tools, highlighting their potential as support agents for reflection and communicative responsibility.

Deploying the proposed framework in real-world scenarios presents inherent challenges. First, relying on LLMs introduces a critical trade-off between computational capacity and response latency [3], particularly in real-time communication channels such as live stream chats. Additionally, existing models often struggle to adequately rephrase implicit or highly contextualized hate speech that requires deep cross-cultural understanding [18]. Additionally, from a behavioral perspective, users may develop desensitization to recurring nudges over time. Mitigating these limitations demands context-aware filtering strategies and dynamic adaptations of the choice architecture to maintain intervention effectiveness.

Finally, empirical results should allow evaluating the plausibility of the behavioral conjectures underpinning the artifact’s design, contributing to the theoretical refinement of models articulating nudging, technology acceptance, and linguistic behavior in digital environments. These findings may serve as a basis for future research investigating similar interventions in other sensitive sociotechnical domains.

VII. CONCLUSION

This paper proposed a study investigating the use of LLM-mediated digital nudges for hate speech mitigation in online writing contexts. Based on the limitations identified in the TD literature, especially regarding user passivity and the risks of perceived algorithmic censorship, the research proposes a reframing of the problem as an assisted decision-making process, in which the user maintains control over their linguistic choices.

The research is structured according to the DSR-Model, articulating knowledge about the problem and context, theoretically grounded behavioral conjectures, the development of a sociotechnical artifact, and a small-scale empirical evaluation. The proposed artifact, materialized in an LLM-mediated Nudge-Based Text Detoxification framework, integrates natural language processing techniques with nudging and HCI principles, seeking to balance technical efficacy, user autonomy, and social acceptability.

The planned empirical evaluation, based on the interaction of real users with a functional prototype, will allow investigating both the artifact’s acceptance and the adequacy of the behavioral conjectures that guided its design. This evaluation is methodologically adequate and sufficient to produce relevant empirical evidence within the context of Design Science Research.

It is expected that the research results will contribute to the advancement of the state of the art in TD systems and to the theoretical deepening of digital nudging and proactive content moderation. By emphasizing the active user role and the importance of non-coercive interventions, the research seeks to offer insights for the development of more transparent, acceptable moderation systems aligned with democratic

values, paving the way for future investigations in similar sociotechnical contexts.

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